Design Thinking for AI Product Prototypes Masterclass

Day 2 Deliverable: Ideation & AI Opportunity Mapping

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Domain: Healthcare 🏥

Chosen Target User Group: Hospital Nurse Managing Patient Records

1. Selected Solution Idea – Hospital Nurse Managing Patient Records

01

Solution Concept

- AI-Powered Smart Documentation Assistant for nurses
- Automatically summarizes patient vitals, medication updates, and progress notes from multiple sources into one structured record

02

How It Solves the Problem

- Reduces repetitive data entry across different systems
- Minimizes time spent on manual record updates
- Ensures information is consistent and accurate

03

User Impact

- Frees up more time for direct patient care
- Reduces stress and workload related to documentation
- Improves efficiency and confidence in patient record accuracy

2. Chosen AI Capability – Hospital Nurse Managing Patient Records

1 AI Capability Type

✅**Natural Language Processing (NLP) - text understanding, chat, summarisation**

❌Machine Learning (prediction, classification)

❌Computer Vision (image/video analysis)

2 AI Task in the Solution

- Automatically reads and summarizes patient notes, vitals, and updates from multiple sources
- Converts unstructured text into structured, accurate entries in electronic health records

3 Why AI is Needed

- Traditional software cannot understand or summarize free-text notes from doctors and nurses
- Manual entry is time-consuming and prone to errors
- NLP enables fast, consistent, and automated record updates, reducing nurse workload and improving accuracy

3. AI Feasibility Map – Hospital Nurse Managing Patient Records

AI FEASIBILITY MAP

1. Data Availability

- **Data Required:** Patient notes, vitals, medication records, progress updates
- **Existing Data:** Most hospitals already collect this digitally in EHR systems
- **Concerns:** Ensure patient privacy (HIPAA compliance), data quality, and completeness

2. Technical Feasibility

- **AI Maturity:** NLP is proven and widely used in healthcare documentation
- **Implementation Complexity:** Medium; requires integration with EHR systems
- **Required Systems:** Hospital EHR software, secure servers, NLP model

3. Operational Feasibility

- **Deployment:** Can be integrated into nurses' existing workflow via EHR interface
- **Maintenance:** Periodic model updates and IT support required
- **Scalability:** Scales across hospital wards or multiple facilities

4. Risk & Constraints

- **Ethical/Regulatory:** Must protect patient confidentiality and follow healthcare regulations
- **Accuracy Limitations:** NLP may misinterpret ambiguous notes; review needed
- **User Adoption:** Nurses need training and trust in AI suggestions for consistent use

4. Justification – Hospital Nurse Managing Patient Records

Why the Solution Fits the Nurse's Problem

- Directly addresses excessive time spent on manual patient record entry
- Reduces repetitive documentation tasks, allowing nurses to focus on patient care

Why AI Adds Real Value

- NLP automates summarization of unstructured notes, improving efficiency
- Ensures accuracy and consistency across multiple patient records
- Frees nurses from repetitive work while maintaining data quality

Why the Solution is Practical Today

- Hospitals already collect digital patient data in EHR systems
- Proven NLP technology is mature and integrates with existing workflows
- Scalable and maintainable without major infrastructure changes